

Complete Fixes Summary

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Config.py

==> (line 23): raised epoch for better training

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data_loader.py

==> (line 38): wrong normalization values (negative mean tiny std) replaced with official CIFAR-10 mean and std values.

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models.py

==> (line 25): (MLP) model used incorrect input size
CIFAR-10 images are of size $3 \times 32 \times 32 = 3072$ so input size can not be 1024

==> (line 40): (MLP) ReLU activation after the final Linear layer of the MLP is mathematically incorrect because CrossEntropyLoss requires raw logits. ReLU clips negative values to zero, distorts the logits distribution, prevents proper gradient backpropagation, and significantly reduces model accuracy. This caused unstable training and poor performance.

==> (line 59 onwards): (CNN) added dropout layer to the CNN architecture it was missing according to readme file.

==> (line 89): (CNN) Fixed input size Input: $3 * 32 * 32$
Block 1 MaxPool $\rightarrow 32$, channels= $16 * 16$
Block 2 MaxPool $\rightarrow 64$, channels= $8 * 8$
So flattened size = $64 * 8 * 8 = 4096$

==> (line 91): (CNN) missing activation function moved to its correct place. ReLU comes before Dropout because Dropout should randomly remove the *activated* features, not the raw linear outputs, to achieve proper regularization.

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train.py

==> (line 83): we will use CrossEntropyLoss instead of NLLLoss
CrossEntropyLoss internally does log_softmax + NLLLoss

==> (line 176): previous code was using train_loader to evaluate the accuracy, which gave fake accuracy because the model has already seen the train data. So instead we should use test_loader for a real accuracy calculation.

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utils.py

==> (line 31): set the model to training mode, this line was missing.

==> (line 48): Gradients accumulate across batches. Clear gradients before backward pass and add optimizer.